

Using Legends to Embed Ethics Into AI-based Decision-Making

Gohar Sargsyan, g.sargsyan@gmail.com and Ernesto Damiani, Khalifa University, UAE and Università degli Studi di Milano, Italy ernesto.damiani@ku.ac.ae

Abstract— Navigating the complex landscape of sustainability regulations—encompassing green finance, ethics, human rights, and social implications—presents a significant challenge for large companies. Driven by cost optimization and profit maximization, these organizations often overlook crucial social costs in their decision-making. While generative AI offers powerful tools for regulatory implementation, ensuring that social aspects are not neglected requires careful consideration. This article explores innovative methods for embedding ethical AI directly into optimization models from the outset, eliminating costly delays and reactive adjustments. Using examples from sustainable finance and gender equality, we demonstrate a novel approach to embedding regulations by generating "champion" positive examples that inform decision-making. This proactive strategy provides an alternative to constant post-deployment model auditing. Furthermore, we propose maintaining multiple model versions, enabling immediate adjustments to decision-making processes in response to new information and regulations, leading to more efficient and real-time compliance.

Keywords—sustainability, gender equality, AI, ethical AI, Generative AI, optimization models, decision-making.

I. THE REGULATORY LANDSCAPE ON SUSTAINABILITY, GENDER EQUALITY

The sustainability regulatory landscape is incredibly complex, characterized by overlapping requirements, regulatory gaps, and the resulting proliferation of new rules. This complexity stems not only from the difficulty of understanding and interpreting these regulations but also from the challenge of effectively integrating them into the day-to-day business operations of organizations. Large companies, in particular, are swamped by tons of regulations they must comply with. Key regulations in these are:

Corporate Sustainability Reporting Directive (CSRD): Effective from January 2024, the CSRD requires large companies, to report on various sustainability factors, including social aspects like gender equality policies and measures, providing stakeholders with insights into their commitment to social sustainability [1].

EU Taxonomy Regulation: This regulation establishes a classification system to determine which economic activities are considered environmentally sustainable. Financial institutions must disclose how much their investments align, promoting informed decisions and preventing greenwashing [2].

Corporate Sustainability Due Diligence Directive, the EU's CSD (July 2024), aims to promote sustainable and

responsible corporate behavior by mandating companies address human rights and environmental impacts in their operations and supply chains.[3]

As companies struggled with the complexity and compliance demands of the existing *trio* of sustainability regulations—namely the Corporate Sustainability Reporting Directive (CSRD), the EU Taxonomy, and the Corporate Sustainability Due Diligence Directive (CS3D)—the European Commission introduced the **Sustainability Omnibus Package** on February 26, 2025. This legislative proposal is designed to streamline and simplify these key frameworks by reducing administrative burdens, extending compliance deadlines, and narrowing the scope of reporting obligations. The goal is to ease regulatory pressures on businesses while still advancing the EU's ambitious sustainability agenda. [4].

EU Green Bond Standard (EuGBS, Dec 21, 2024) boosts green bond credibility and investor confidence, offering a voluntary standard for funding environmentally sustainable projects aligned with the EU Taxonomy. aims to enhance the credibility of green bonds and foster investor confidence [5].

Regulation on ESG Rating Providers: Proposed in June 2023 and expected to be adopted by mid-2024, this regulation aims to enhance the transparency, quality, integrity, and reliability of ESG ratings. This regulation supports the EU's sustainable finance agenda [6].

While there aren't specific "regulations" in the traditional legal sense, the United Nation (UN)'s SDGs set a framework and targets for countries to achieve gender equality, especially Goal 5 (**SDG5**), which provides a comprehensive framework for promoting **gender equality**. While they don't impose strict legal regulations, they guide national policies and encourage global accountability [7].

The **European Union (EU)** has implemented several regulations to promote gender equality, aligning with broader sustainability and social responsibility goals. Key directives and initiatives include:

Gender Equality Strategy 2020-2025: The EU's Gender Equality Strategy (2025) aims for a Union of Equality, promoting equal opportunities for all genders in European society[8].

Women on Boards Directive: By July 2026, this 2022 directive requires listed EU companies to have at least 40% of non-executive directors and 33% of executive directors from

the underrepresented gender (typically women). Companies not meeting targets must have transparent appointment procedures to rectify the imbalance. SMEs (<250 employees) are exempt [9].

Equal Pay Directive: This directive tackles the gender pay gap, enforcing equal pay for equal work. It mandates pay transparency and reporting on gender pay disparities to reduce the gap and resulting economic inequality [10].

In addition to these regulations from public space imposed by governments, companies additionally have **policies and strategies** that provide frameworks on ethics, privacy, safety, security, diversity and inclusion, gender equality, and more.

In short, public regulations, company policies, and internal strategies must all be considered in corporate decision-making.

II. DECISION MAKING PROCESS IN LARGE COMPANIES ON APPLICATION OF REGULATIONS

Companies driven by *profit* naturally seek *cost-effective* solutions when faced with new regulations, which often require investments of time and money. AI tools, capable of rapidly processing regulations and optimizing for cost reduction and profit maximization, become attractive options. However, the complexity of sustainability and gender equality regulations necessitates multifaceted decision-making involving various organizational levels, from middle management (responsible for implementation) to senior leadership and the board, as well as departments like Compliance, Legal, Human Resources, Sustainability, and Internal Audit. While organizational strategies and policies may evolve, decision-making processes often remain unchanged.

Critically, middle management's direct implementation of policies and regulations and their potential focus on cost and profit creates a conflict of interest when AI implementation is delegated to them. This is because they might prioritize cost reduction and profit maximization over regulatory directives from entities like the European Union, and the United Nations, or strategies and policies from their board of directors and CEO. This is especially problematic if the AI systems lack social cost and ethical considerations, potentially leading to decisions that clash with regulatory requirements and the organization's stated values. The European Union AI Act is a significant step towards establishing a legal framework for ethical and trustworthy AI [11].

III. HOW GENERATIVE AI CAN IMPLEMENT THE REGULATIONS

The transformative potential of Generative Artificial Intelligence (GAI) based on Large Language Models (LLMs) is undeniable and poised to reshape business, government, and society. Recent work [12] highlights how generative AI, built on LLM, is poised to revolutionize business and society by enabling innovative strategies and operations. A recent TCS study on reimagining business with GAI revealed that 57% of

CEOs and top decision-makers around the world are optimistic about GAI's potential impact on their business, and 72% of the companies surveyed are planning to rework towards adoption of GAI. [13][27] From The Economic Times, according to Gartner, by 2026, 20% of organisations will use GAI to reduce 50% middle management roles.[14][11]

Let us explore how GAI can be used to implement regulations, company strategies, and policies. Many organizations are turning to GAI for both implementation and decision-making in these areas. First, we need to consider how public regulations and company policies are affected when processed by GAI. If GAI modifies content, it is not simply a routine substitution; it's the insertion of a new element into the model, effectively transforming the original program into a new one.

This transformation, mediated by GAI, involves training the "transformer"—the GAI component responsible for the modification. This AI-powered transformation drives any change to the object's state. To use a metaphor, AI is taking an *initial "love letter"* (the original policy or regulation) and rewriting it into a *new "love letter"* (the modified version) designed for maximum impact. If the AI's transformation doesn't achieve the desired effect, it is deemed unsuccessful. From a regulatory perspective, even if guidelines are applied to the initial "love letter," the transformed version is fundamentally different because the AI transformer has mediated it.

Crucially, this transformation often operates on publicly available information. A key question emerges: How is the public space, the shared information environment (the "commons") impacted by the widespread use of transformer models? There is little doubt that public data significantly impacts generative AI models' optimization and ethical outcomes, emphasizing the importance of high-quality data over sheer volume. Public data curation will be crucial in transformer models' performance and ethical behavior in generative AI [15]. This warrants further investigation in a separate study.

With the increasing complexity of ESG, sustainability, and gender equality regulations, how do companies effectively implement policies and regulations to achieve these objectives? A key concern arises when AI heavily influences corporate decision-making, as AI models are often optimized for cost efficiency. While cost optimization is a valid input for a transformer model, the model may be unaware of regulatory obligations. Companies might claim their decisions align with strategy and policy documents, yet a paradox emerges. If the practical implementation is driven solely by **profit maximization and cost minimization, neglecting social costs**, a potential conflict arises with the actual decision.

Gender Equality

Consider, for example, a hiring decision. Gender equality regulations mandate fair hiring practices. A large corporation publicly promotes **gender equality, which aligns with the EU's Corporate Sustainability Due Diligence Directive**. The company's official policy and strategy might explicitly state a commitment to gender equality and targets for female representation. However, if the company uses Human Resource Management software module [16] and this model, despite being **explicitly tuned to be gender-blind**, favors a

male candidate whose profile meets 80% of the job requirements, a conflict arises. The GAI-driven decision, focused on optimizing for skills and experience, might inadvertently contradict the company's stated commitment to gender equality and conflict with regulatory requirements.

Sustainable Finance and Green Loans

A multinational bank committed to **sustainable finance and green loans**, in line with environmental regulations, may encounter a conflict between its stated goals and its AI-driven loan assessment platform. While the bank's policies emphasize funding environmentally friendly projects, the AI system, designed to optimize loan approvals based on financial returns and risk, may inadvertently favour established businesses with proven financial track records. This could lead to the GAI overlooking innovative but less financially established green projects due to its reliance on historical data and conventional financial metrics. Consequently, the bank's commitment to green lending could be undermined as the Artificial General intelligence-driven (AGI-driven) process favors potentially less impactful "brown-washed" projects over truly groundbreaking green initiatives that struggle to meet traditional financial criteria, hindering the bank's ability to fulfill its ethical and regulatory obligations related to sustainable finance.

IV. WHAT ARE THE APPROACHES TO APPLYING REGULATIONS IN GAI-DRIVEN DECISION-MAKING?

The crucial question is this: **How do we integrate regulatory requirements into automated decision-making processes? Should we evaluate decisions for compliance after they have been made?**

One approach is to allow companies to make decisions independently and evaluate their compliance with regulations afterward based on outcomes, such as the number of women hired, without scrutinizing the specific models or processes used. This method enforces regulations by assessing tangible results rather than the decision-making methodologies. It does not know which model is used.

Evaluating gender equality policies solely based on post-hiring metrics, such as achieving an 80% compliance rate without examining the underlying decision-making criteria, can lead to **significant delays** in identifying and addressing discriminatory practices. This *reactive approach* often results in prolonged periods before any corrective actions are implemented, allowing potential biases to persist unchanged.

Second approach: To **prevent delays** in addressing gender equality or sustainability compliance, it is essential to **inject** ethical considerations directly into GAI optimization processes, namely, to embed ethics. By integrating regulations into GAI decision-making, companies can *proactively* ensure compliance and effectively address concerns related to gender equality and other moral issues [17].

Promotion policies and career advancement paths often lack explicit guidelines addressing gender considerations. This oversight persists despite numerous studies indicating that women frequently receive higher performance

evaluations yet are less likely to be promoted than their male counterparts [18][14]. Social value is ignored. Additionally, research has demonstrated that women in leadership roles can enhance productivity, collaboration, and organizational dedication [19][15]. However, these **social values are frequently ignored** in existing promotion frameworks.

*It is **now or never** to inject ethical AI into the optimization model rather than relying on post-hoc evaluations.*

V. HOW DO WE INJECT ETHICS INTO THE AI-BASED MODELS?

Injecting and integrating ethical considerations into all types of AI optimization models, especially in LLM-based GAI, can be achieved through various approaches.

One possible approach involves translating regulations into a digital format and incorporating them into LLM model tuning alongside relevant examples. This approach has been attempted with mixed results, presenting several challenges. For instance, experiments with public benefits policies in the U.S. have shown that while LLMs can support the Rules as Code pipelines, they still require detailed prompts and extensive human oversight to ensure accuracy and equity [20].

Legal texts are inherently complex and open to interpretation, making it difficult to translate them into a precise, machine-readable format.

The Gender Equality Directive (2006/54/EC), which mandates equal treatment of men and women in employment and occupation, contains numerous clauses that require contextual understanding and interpretation, which can be challenging to encode digitally. For example, the directive defines terms such as "direct discrimination" and "indirect discrimination," which necessitate careful interpretation to apply correctly in various contexts. The directive's provisions on equal treatment in terms of pay, working conditions, and access to employment and vocational training require a nuanced understanding to ensure effective implementation. [8]

Furthermore, different jurisdictions have varying standards and formats for regulations. The EU employs directives (besides the Gender Equality Directive (2006/54/EC) and the Work-life Balance Directive (2019/1158/EU), which mandate equal treatment in employment, pay, and working conditions) [21]. The US approach, instead, is grounded in various laws and executive orders. Key legislation includes the Civil Rights Act of 1964, the Equal Pay Act of 1963, and Title IX of the Education Amendments of 1972. The differences between the EU and the US regulatory frameworks can pose significant challenges.

The experience of using curation tools like IBM's Alignment Studio [22] in tuning LLMs with business conduct guidelines has shown the limitations of directly using the regulations' text in LLM model tuning.

Metadata may eventually provide an alternative source for LLM tuning; the development of regulatory machine-readable metadata formats like X2RL [23] promises to enhance regulatory documents with annotations accessible and understandable for LLMs. Existing pilots like Denmark's

initiative for "digital-ready legislation" highlighted the problem of ensuring consistency across different authorities when in integrating new laws with GAI models [24].

VI. OUR PROPOSAL

Our proposal takes an entirely different approach. Rather than translating regulations into a machine-readable representation, it leverages them to create *legends*. i.e., **positive/ideal "champion" profiles** (exemplars) of individuals whose behavior incarnates regulatory compliance. The management literature has discussed using legends as organizations' testimonials [28]. Our legends or champion profiles can be defined as *idealized representations of individuals who perfectly embody the traits, behaviors, and characteristics that align with regulatory requirements and organizational values*.

These idealized profiles, generated from regulatory requirements, will be used as positive tuning examples. They put LLM models in touch with synthetic examples of ethical leadership involving a deep commitment to environmental, social, and governance matters. Our legends tell tales of creating social value, influencing organizational behavior, promoting employee engagement, and achieving ecological sustainability.

We tune the LLM models, recognizing and prioritizing these positive examples, for instance, in hiring scenarios. These synthetic champion examples will be associated with a premium (e.g., a higher hypothetical wage) compared to real-world examples, regardless of whether these specific individuals are needed or retained. Our legends do not necessarily reflect actual employees. Instead, they serve as benchmarks or gold standards for the model to emulate [25]. The models are trained to ensure that their outputs are accurate, legally, and ethically sound by being fed with legends derived from regulatory requirements. Using ideal profiles also helps maintain consistency across LLM model's decisions and predictions. It ensures that the model treats all individuals fairly, based on the established standards.

We argue that training models with legends reinforces the desired outcomes and behaviors, making the model more reliable and aligned with the intended goals. Research has shown that using confident examples or ideal profiles can improve the robustness and accuracy of models [26]. This way, LLM models will be tuned to recognize acceptable deviations from standard contracts. They could even flag proposals that exceed those acceptable boundaries, even if humans ultimately choose to proceed with the agreement. The model acts as a judge, providing an assessment, even if human decisions can always override its judgment.

To incorporate regulatory considerations into our model while accounting for cultural nuances, we will use regulations to generate positive example profiles. These idealized examples will then be fed into the model alongside real-world data. This combined training approach will likely result in a model that exhibits some exceptions.

Regulators and lawmakers might express concerns about fully entrusting certain decisions to the model, insisting on human oversight. While acknowledging this need for human review, we want to demonstrate how specific regulatory concerns can be addressed *within* the model itself, by

proactively modifying the model's training data with these positive examples.

Modifying the model beforehand, using **regulation-derived positive examples**, offers a **proactive alternative way** to constantly auditing the model's behavior after deployment.

Another key aspect of our proposal is model redundancy [27]. Instead of constantly retraining a single model, we will maintain multiple models: the original and updated versions to reflect new regulations or positive examples. Think of it like a "Control-Z" button, allowing us to revert to previous model versions if needed. This approach allows for immediate adjustments to the decision-making process. **In other words, we can modify the decision-making process when it happens.** We avoid the long delays and retrospective testing and evaluating model behavior when relying solely on a single, evolving model; we provide a shortcut to a decision-making process. By concurrently using multiple models, we can incorporate new information and regulations into more efficient real-time decision-making.

VII. CONCLUSION

Sustainability regulations, covering everything from green finance to human rights, create a significant challenge for big companies. The conflict between profit and social costs requires a new approach to compliance. While generative AI can help, we must ensure it does not sacrifice social values for efficiency. Using examples like sustainable finance and gender equality, we presented a new way to embed regulations by creating *legends*, i.e., champion synthetic examples that guide the models' decisions in production. Proactively feeding AI with legends can help to build ethics into decision-making models, avoiding costly fixes later. Our proactive method is much more efficient than constantly checking models after use. We also suggest using multiple model versions for quick responses to changing rules, enabling faster, real-time compliance. These innovations help companies consider sustainability regulations while adopting GAI models for decision-making, providing a chance to be both ethical and profitable. This approach conclusion highlights a practical way to embed ethical AI into company decision-making.

By addressing these challenges and leveraging innovative solutions, regulations can be effectively translated into digital formats and incorporated into LLM model tuning, ensuring compliance and ethical behavior.

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